

Practice Sheet 1.2: Basics of Python Programming

11. Circle Area and Circumference

Problem Statement:

Write a Python program to calculate the **area** and **circumference** of a circle using the given radius.

Assume $\pi = 3.14$.

Input Format:

Radius of the circle (float)

Output Format:

Print the area and circumference.

12. Total Working Hours

Problem Statement:

A company records working time in hours and minutes. Write a Python program to calculate the **total working hours in decimal format**.

Input Format:

Hours (int)

Minutes (int)

Output Format:

Print total working hours.

13. Convert Days into Years

Problem Statement:

Write a Python program to convert the given number of **days into years**. Assume 1 year = 365 days.

Input Format:

Number of days (int)

Output Format:

Print number of years.

14. Perimeter of a Square

Problem Statement:

Write a Python program to calculate the **perimeter of a square** using the given side length.

Input Format:

Side of the square (int)

Output Format:

Print the perimeter.

15. Total and Average of Five Numbers

Problem Statement:

Write a Python program to calculate the **total** and **average** of five numbers entered by the user.

Input Format:

Five numbers (float)

Output Format:

Print the total and average.

16. Electricity Bill Calculation

Problem Statement:

Write a Python program to calculate the **electricity bill** by multiplying the total units consumed with the cost per unit.

Input Format:

Units consumed (int)

Cost per unit (float)

Output Format:

Print the electricity bill amount.

17. Convert Kilometers to Meters

Problem Statement:

Write a Python program to convert the given distance from **kilometers to meters**.

Input Format:

Distance in kilometers (float)

Output Format:

Print distance in meters.

18. Discount Calculation

Problem Statement:

A shop offers a **flat discount amount** on a product. Write a Python program to calculate the **final price** after discount.

Input Format:

Original price (float)

Discount amount (float)

Output Format:

Print final price.

19. Calculate Speed

Problem Statement:

Write a Python program to calculate the **speed** of a vehicle using the formula:
 $\text{Speed} = \text{Distance} / \text{Time}$

Input Format:

Distance (float)

Time (float)

Output Format:

Print the speed.

20. Convert Grams to Kilograms

Problem Statement:

Write a Python program to convert the given weight from **grams to kilograms**.

Input Format:

Weight in grams (float)

Output Format:

Print weight in kilograms.